Stats for Sociologists

Course: SOC6302H1F

Instructor: Professor Pepin

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INSTRUCTIONS:

Fill out the first page of the scantron. Confirm you have included:

- Your student number (write the number and fill in the ovals)
- Your surname and first initial (write the letters and fill in the ovals)

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1. Which of the following best describes the differences between the Console, R Scripts, and Quarto documents in RStudio?
   (a) The Console runs code interactively, R Scripts store reusable code, and Quarto combines code, text, and output for dynamic reports.
   (b) The Console and R Scripts are identical, while Quarto is used only for Markdown formatting.
   (c) The Console is used for writing long-term code, R Scripts are for temporary testing, and Quarto is only for plotting.
   (d) R Scripts and Quarto documents both run code line-by-line, but only the Console saves your work.

2. Based on the table, which description best describes the correlation between personal income (`realrinc`) and number of days of poor mental health (`mntlhlth`)?

Correlation Coefficient	t-statistic	df	p-value	95% CI Lower	95% CI Upper
-0.114	-5.04	1933	0.000	-0.16	-0.07

   (a) There is a weakly positive association between personal income and days of poor mental health ($r = -0.11$, $p < .05$).
   (b) There is no evidence of an association between personal income and days of poor mental health ($t = -5$).
   (c) Decreases in personal income was correlated with decreases in the number of days of poor mental health ($t = -5$, $p < .05$).
   (d) Results indicate an inverse relationship ($r = -0.11$, $p < .05$), suggesting that the number of days of poor mental health decreases as personal income increases.

3. According to the 68-95-99.7 rule, what can we say about a person whose score is more than 2 standard deviations above the mean?
   (a) Their score is within the middle 68% of the distribution.
   (b) Their score is likely below average.
   (c) Their score is very common.
   (d) Their score is unusual and higher than most.

4. According to Cohen's argument in *Citizen Scholar*, what role should descriptive statistics play in social science research?
   (a) Descriptive statistics are a foundational part of scientific inquiry and should be valued as legitimate research.
   (b) Descriptive statistics are useful predominately as a step before hypothesis testing and theory development.
   (c) Descriptive work should be minimized in favr of studies that make strong theoretical contributions.
   (d) Descriptive statistics are outdated and should be replaced by more advanced inferential methods.

5. If the final grades in a statistics class were normally distributed with an average of 81.5% and a standard deviation of 1.5, about what proportion of the students' final grades were between 80% and 83%?

   (a) 50%
   (b) 62%
   (c) 68%
   (d) 95%

6. In a normally distributed set of test scores, a student scoring below the average will have a _____.

   (a) negative z-score
   (b) negative standard error
   (c) positive standard error
   (d) positive z-score

7. Which of the following is a measure of the spread of a distribution?

   (a) skewness
   (b) standard deviation
   (c) sample size
   (d) standardized variable

8. Researchers calculated a 95% confidence interval of the average family size of Canadians to be from 2.7 to 3.1, based on a sample of 500 families with a standard deviation of 1.1. What change would produce a more precise estimate?

   (a) Increasing our standard deviation to 1.3
   (b) Increasing our confidence interval to 99%
   (c) Increasing our sample size to 600
   (d) Improving our sampling strategy

9. In *White Logic, White Methods*, what does Gandy suggest about the use of racial statistics?

   (a) Their use in data journalism helps mitigate online polarization
   (b) They are neutral tools most useful for academic research
   (c) Their use in legal proceedings increasingly lead to unfair policy outcomes
   (d) They can both expose and perpetuate discrimination

10. Which of the following code snippets correctly uses `mutate()` to convert height from centimeters to meters?

    (a) mutate(height / 100 = height)
    (b) mutate(height <- height / 100)
    (c) mutate(height = height / 100)
    (d) mutate(height = select(height / 100))

11. Wheelan illustrates what key concept in the chapter "Inference: Why My Statistics Professor Thought I Might Have Cheated?" from the book *Naked Statistics: Stripping the Dread from the Data*?

    (a) That cheating can be detected by comparing mean scores across different groups
    (b) That hypothesis testing is not useful in criminal investigations
    (c) That statistical inference can be used to prove someone's guilt beyond reasonable doubt
    (d) That unusually accurate results can raise suspicion, but statistical inference helps assess whether they are likely due to chance

12. A sociologist wants to examine whether there is a relationship between level of education (measured in years) and annual income (in dollars) among adults in the U.S. Which statistical test is most appropriate?

    (a) Logistic regression
    (b) t-test
    (c) Chi-square test
    (d) Pearson's correlation (r)

13. A researcher summarized the `sibs` variable from the U.S. General Social Survey, a measure of how many brothers and sisters respondents had. Based on the summary table, the distribution appears to be:

freq	median	mean	sd
3962	3	2.9	1.8

    (a) negatively skewed
    (b) nominal
    (c) positively skewed
    (d) symmetrical

14. Which description best describes a correlation coefficient of r = -0.62?

    (a) Weakly negative relationship
    (b) Strong negative relationship
    (c) A statistically insignificant relationship
    (d) No relationship exists between the variables

15. Which RStudio pane is used to write and edit R scripts and Quarto documents?

    (a) Files
    (b) Source
    (c) Environment
    (d) Console

16. Which of the following is a valid use of `group_by()` and `summarize()`?

    (a) data |> group_by(sex) |> summarize(freq = n())
    (b) data |> summarize(mean = group_by(sex))
    (c) summarize(group_by(data, age), mean(sex))
    (d) group_by(data, summarize(sex))

17. Researchers found that couples report about 10 arguments per month on average. If a couple received a z-score of -2 for this measure, which of the following best describes the amount of arguments for this couple compared with the average?

    (a) This couple had fewer arguments than about 98% of couples.
    (b) This couple had about 8 arguments per month.
    (c) This couple had fewer arguments than about 5% of couples.
    (d) This couple had fewer arguments than about 2% of couples.

18. Using 2024 GSS data, a researcher estimated the hours per day Americans spend relaxing. Based on the 95% confidence interval from R, which interpretation is correct?

sex	n	mean	sd	lowCI	hiCI
male	1139	4.2	3.7	4.0	4.5
female	1174	4.0	3.5	3.8	4.2

    (a) We cannot determine whether the null hypothesis of zero difference between the parameters will be rejected at the 0.05 level.
    (b) The confidence intervals do not overlap so we can conclude that the average time men and women spend on leisure activities per day are statistically significantly different.
    (c) We are 95% certain that women have less time per day than men for leisure activities, likely because of their greater time spent on housework and childcare.
    (d) The confidence intervals overlap, so we can't know for sure if there is a statistical difference between the average time men and women spend on leisure activities per day.

19. If a researcher concluded that there was no difference in police use of force against Black and White men when there actually is a difference, then:

    (a) The null hypothesis was correctly retained.
    (b) The null hypothesis was correctly rejected.
    (c) The null hypothesis was incorrectly retained (false negative).
    (d) The null hypothesis was incorrectly rejected (false positive).

20. What does the `::` operator do in R?

    (a) Imports all functions from a package
    (b) Comments out a line of code
    (c) Specifies a function from a particular package
    (d) Assigns a value to a variable

21. Based on the table, which statement best compares the relative frequency of being 'not too happy' between men and women?

sex	male	female	Total
happy			
very happy	350 (19.8%)	441 (20.5%)	791 (20.1%)
pretty happy	1025 (57.8%)	1264 (58.7%)	2289 (58.3%)
not too happy	397 (22.4%)	450 (20.9%)	847 (21.6%)
Total	1772 (100.0%)	2155 (100.0%)	3927 (100.0%)

(a) A greater number of women reported being 'not too happy' than men, so women have a higher relative frequency.
(b) Men have a slightly higher relative frequency of being 'not too happy' than women, even though fewer men reported it.
(c) Women have a slightly higher relative frequency because more women than men reported being 'not too happy'.
(d) The frequency of being 'not too happy' is equal for men and women.

22. A respondent reports 5 poor mental health days in the past 30 days. Based on summary stats for the `mntlhlth` variable, which of the following best describes their z-score?

freq	median	mean	sd
2279	0	4.5	7.9

(a) The z-score is negative because 5 is below the sd.
(b) The z-score is large and positive because 5 is much higher than the mean.
(c) The z-score is close to zero because 5 is close to the mean.
(d) The z-score is undefined because the median is 0.

23. What is the result of the following code?

```
gss24 |>
  filter(age > 20) |>
  select(age, childs)
```

(a) The original gss24 dataset is overwritten.
(b) A new column is added to `gss24`
(c) rows with `age` > 20 are removed
(d) Only rows with `age` > 20 and columns `age` and `childs` are returned.

24. In the 1970s, Americans watched an average of 3.05 TV hours per day, compared with an average of 3.3 hours per day in 2024. Assuming an α level of .05, what does the researcher

t-statistic	df	p-value	95% CI Lower	95% CI Upper
-2.84	3616.79	0.005	-0.43	-0.08

conclude about the statistically significant difference?

(a) The negative t-statistic suggests a decrease, meaning Americans watch less TV in 2024 than in the 1970s.
(b) Although there is a statistically significant difference, the direction of change is unclear.
(c) The means indicate that Americans watch more TV in 2024 than in the 1970s.
(d) The confidence interval is below zero, indicating Americans watch less TV in 2024 than in the 1970s.

25. What will the following code do?

```
gss24 <- gss24 |>
  mutate(status = case_when(
  income >= 100000 ~ "High",
  income > 50000 & income < 100000 ~ "Medium",
  TRUE ~ "Low"
))
```

(a) It filters out rows where income is less than 50,000.
(b) It creates a new column named `status` with 3 possible values.
(c) It replaces the numeric values in the income column with text labels.
(d) It removes missing values from the income column.

26. According to Wheelan, in *Naked Statistics: Stripping the Dread from the Data?*, what is the most common cause of inaccurate polling results?

(a) Incorrect calculations of standard errors
(b) Random variation in respondent answers
(c) Misuse of statistical software during data analysis
(d) Biased sampling methods or poorly worded questions

27. What is Wheelan's primary message of Chapter 1 ("What's the Point?") in *Naked Statistics: Stripping the Dread from the Data?*

(a) Statistics explain why relationship exist between variables
(b) Statistics provide honest answers to complex questions
(c) Statistical formulas are the most important part of understanding data
(d) Statistics simplify complex information to help us make better decisions

28. Has agreement with the statement that 'Black Americans face worse jobs, income, and housing than white Americans due to discrimination' shifted? Using a significance level of $\alpha = 0.01$, what conclusion should be drawn?

	year		
	2022	2024	Total
opinion			
no	509 (45%)	522 (51%)	1,031 (48%)
yes	619 (55%)	494 (49%)	1,113 (52%)
Total	1,128 (100%)	1,016 (100%)	2,144 (100%)

Pearson's Chi-squared test, p=0.004

(a) The small change is statistically significant, but most people still agree with the statement.
(b) The increase in agreement with the statement between 2022 and 2024 is statistically significant.
(c) The drop in agreement with the statement between 2022 and 2024 is statistically significant.
(d) There is a statistically significant difference, so we fail to reject the null hypothesis.

29. What will happen if you try to use a function from a package that hasn't been loaded?

(a) The function will run normally
(b) R will prompt you to install the package
(c) R will automatically load the package
(d) You will get an error saying the function is not found

30. What does the following code return?

```
gss24 |>
  group_by(sex) |>
  summarize(mean_age = mean(age, na.rm = TRUE))
```

(a) A list of all ages grouped by sex.
(b) A data frame showing the average age for each sex.
(c) A filtered data frame with only rows where age is not missing.
(d) A new column named `mean_age` added to the original data frame.

31. Chi-square values cannot be _____.

(a) zero
(b) negative
(c) positive
(d) outside the range of 0 to 1

32. What does the following R code produce?

```
gss21 |>
  tbl_cross(
    row = incgap,
    col = sex,
```

```
    percent = "column",
    missing = "no") |>
  add_p()
```

(a) A summary table of `sex` grouped by `incgap`, with missing values excluded and a t-test.
(b) A cross-tabulation comparing the mean values of `incgap` between men and women using a t-test
(c) A cross-tabulation of `incgap` by `sex`, showing column percentages and a chi-square test of independence
(d) A frequency table of `incgap` responses with missing values imputed and grouped by `sex`

33. Which of these datasets has a standard deviation of 0?

(a) 4, 4, 4, 4, 4
(b) 1, 4, 7, 10, 15
(c) 0, 2, 4, 6, 8
(d) 2, 2, 5, 7, 7

34. Which command installs a new package in R?

(a) `library()`
(b) `install.packages()`
(c) `load()`
(d) `require()`

35. Why is it considered good practice to use RStudio Projects when working on data analysis?

(a) They automatically load packages when you open RStudio
(b) They ensure your code runs faster by optimizing memory usage
(c) They prevent syntax errors by enforcing tidyverse formatting rules
(d) They help organize your files, set a working directory, and keep your environment consistent

36. National survey data shows that 69% of Canadians support mandatory childhood vaccinations. A local researcher believes that support is likely much higher in Toronto compared with the national average. Which of the following equations represents the research hypothesis?

(a) $\mu \neq 0.69$
(b) $\mu > 0.69$
(c) $\mu < 0.69$
(d) $\mu = 0.69$

37. Using 2024 GSS data, a researcher estimated how many Americans believe in life after death. Based on the 90% confidence interval from R, which interpretation is correct?

n	mean	sd	lowCI	hiCI
1283	0.83	0.38	0.81	0.85

(a) Because the mean is within the confidence interval, there is a .90 probability that the proportion of U.S. adults who believe in some form of life after death is .83.
(b) If another survey had been conducted in 2024, there's a 90% chance the point estimate would be between .81 and .85.
(c) About 90% of people responded to the national survey.
(d) The researcher is 90% confident that the proportion of U.S. adults who believe in some form of life after death is between .81 and .85.

38. A researcher compares support for universal healthcare (Support vs. Oppose) between two racial groups: Black and White respondents. Which statistical test should be used?

(a) Linear regression
(b) Chi-square test
(c) Pearson's correlation (r)
(d) t-test

39. What does the pipe operator (|>) do in dplyr?

(a) It filters rows based on a condition.
(b) It loads the dplyr package into memory.
(c) It creates a new column in a data frame.
(d) It passes the result of one function into the next.

40. When examining a frequency distribution, which of the following columns generally allows you to identify the median most quickly?

(a) cumulative frequencies
(b) cumulative percentages
(c) frequencies
(d) percentages

41. A p-value is

(a) A measure of how unusual or rare our obtained statistic is compared with what is stated in the null hypothesis
(b) An estimate of the standard error of a sampling distribution
(c) An estimate of where our population statistic lies on the true sampling distribution of the means
(d) A measure that tells us the variability in a normal distribution

42. What does the expression `5 != 3` evaluate to in R?

(a) TRUE
(b) NA
(c) NULL
(d) FALSE

43. Using the U.S. General Social Survey data, a researcher tests whether the proportions who favor a law requiring a person to obtain a police permit before they could buy a gun have changed over time. Assuming an α level of .05, which of these statements best summarizes the researcher's conclusion?

t-statistic	df	p-value	95% CI Lower	95% CI Upper
1.82	3114.93	0.069	-0.00	0.06

(a) There is a statistically significant increase in support for the police permit law from 1984 to 2024, proving that stricter gun control laws are now favored by most Americans.
(b) There is a statistically significant difference in support for the police permit law between 1984 and 2024, but it was due to sampling error and cannot be interpreted.
(c) There is a statistically significant difference in support for the police permit law between 1984 and 2024, suggesting that public opinion has changed over time.
(d) There is not a statistically significant difference in support for the police permit law between 1984 and 2024, indicating that public opinion has remained stable.

44. In order to compare meaningfully across groups, a frequency should be converted into a/an _____.

(a) percentage
(b) z-score
(c) raw count
(d) cumulative frequency

45. In *Naked Statistics: Stripping the Dread from the Data?*, Wheelan explains that "most sample means will lie reasonably close to the population mean; the standard error is what defines 'reasonably close.' " What does this statement imply?

(a) The standard error tells us how far the population mean is from the sample mean.
(b) The standard error measures how much sample means tend to vary around the population mean.
(c) The sample mean is always equal to the population mean.
(d) The standard error increases as sample size increases.

46. Using the table, which of the following is the correct formula to find the expected frequency for U.S. liberals who oppose the death penalty in 2024?

ideology	conservative	liberal	Total
cappun			
favor	573 (63.0%)	337 (37.0%)	910 (100.0%)
oppose	204 (32.4%)	425 (67.6%)	629 (100.0%)
Total	777 (50.5%)	762 (49.5%)	1539 (100.0%)

(a) (600 * 513)/1280
(b) (600 * 767)/1280
(c) (345 * 513)/1280
(d) (345 * 600)/1280

47. The level of measurement for the `hapmar` variable is _____

[1] "Happiness of marriage"

	Freq	%	% Cum.
very happy	931	56.60	56.60
pretty happy	646	39.27	95.87
not too happy	68	4.13	100.00
Total	1645	100.00	100.00

(a) nominal
(b) dichotomous
(c) interval
(d) ordinal

48. Which of the following statements best reflects the ASA's guidance on interpreting p-values?

(a) A small p-value always indicates a large and important effect.
(b) A large p-value proves that no effect exists.
(c) p-values are influenced by sample size and measurement precision, and do not reflect effect size or importance.
(d) p-values measure the scientific or economic significance of a result.

49. Which function is used to load a package in R?

(a) `load()`
(b) `source()`
(c) `install.packages()`
(d) `library()`

50. What is the purpose of `drop_na()` in dplyr?

(a) It replaces missing values with zeros.
(b) It drops columns that are entirely NA.
(c) It converts NA values to NULL.
(d) It removes rows that contain missing values.